EVERY SMOOTH COVERING OF AN ALGEBRAIC SPACE IS REFINED BY AN ETALE COVERING

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Surjective etale charts. Spaces, section on presentations (cross-chapter, structural). Supplies U -> X and W_i -> X_i surjective etale from schemes.
- Scheme-level etale dominance of smooth coverings. More on Morphisms, lemma etale-dominates-smooth (cross-chapter, named). Refines smooth coverings of schemes by etale coverings.

Source reference. Original reference 055V: focused-seeds/geometry/S9-topo-etale-dominates-smooth/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let $\{X_i \to X\}_{i \in I}$ be a smooth covering of X. Then there exists an étale covering $\{U_j \to X\}_{j \in J}$ of X which refines $\{X_i \to X\}_{i \in I}$.*

Proof. First choose a scheme U and a surjective étale morphism $U \to X$. For each i choose a scheme W_i and a surjective étale morphism $W_i \to X_i$. Then $\{W_i \to X\}_{i \in I}$ is a smooth covering which refines $\{X_i \to X\}_{i \in I}$. Hence $\{W_i \times_X U \to U\}_{i \in I}$ is a smooth covering of schemes. By More on Morphisms, Lemma 055V we can choose an étale covering $\{U_j \to U\}$ which refines $\{W_i \times_X U \to U\}$. Then $\{U_j \to X\}_{j \in J}$ is an étale covering refining $\{X_i \to X\}_{i \in I}$. □

Date: stacks-project:spaces-topologies.tex:local-snapshot-20261012; GNU FDL 1.2 or later.